

☐ Why study Computer Architecture and Computer Organization?



- Computer architecture is a key component of computer engineering and the practicing computer engineer should have a practical understanding of this topic.
- It is concerned with all aspects of the design and organization of the central processing unit and the integration of the CPU into the computer system itself.

X **Computer architecture** is concerned with X
structure and behavior of computer as seen by
the user .the architectural design of a computer
system concerned with the specification of the
various functional modules ,such as processors
and memories ,and structuring them together in
to a computer system .

Computer organization is concerned with the way
the hardware components operate and the way
they are connected together to from computer
system .

Possible Computer Operation

- ☐ The computer, of course, must be able to **process data**. The data may take a wide variety of forms, and the range of processing requirements is broad .
- ☐ It is also essential that a computer **store data**. If the computer is processing data on the fly (i.e., data come in and get processed, and the results go out immediately), the computer must temporarily store at least those pieces of data that are being worked on at any given moment.



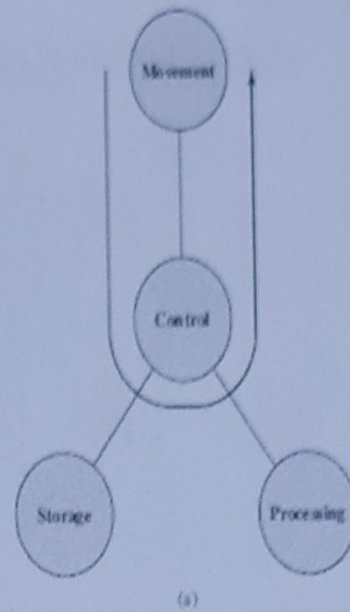
Possible Computer Operation



- Finally, there must be control of these three functions.
- Within the computer , a control unit manages the computer's resources and orchestrates the performance of its functional parts in response to instructions.

Possible Computer Operation

Data movement device
simply transferring data
From one peripheral
or communications
line to another.

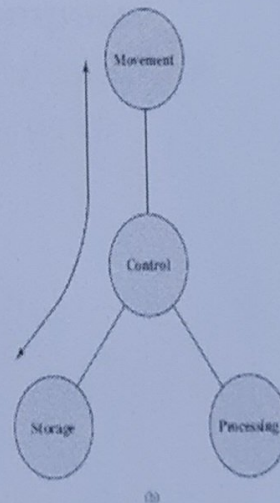


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Possible Computer Operation

function as a **Data storage** device.
with data transferred from
the external environment
to computer storage
(read) & (write).

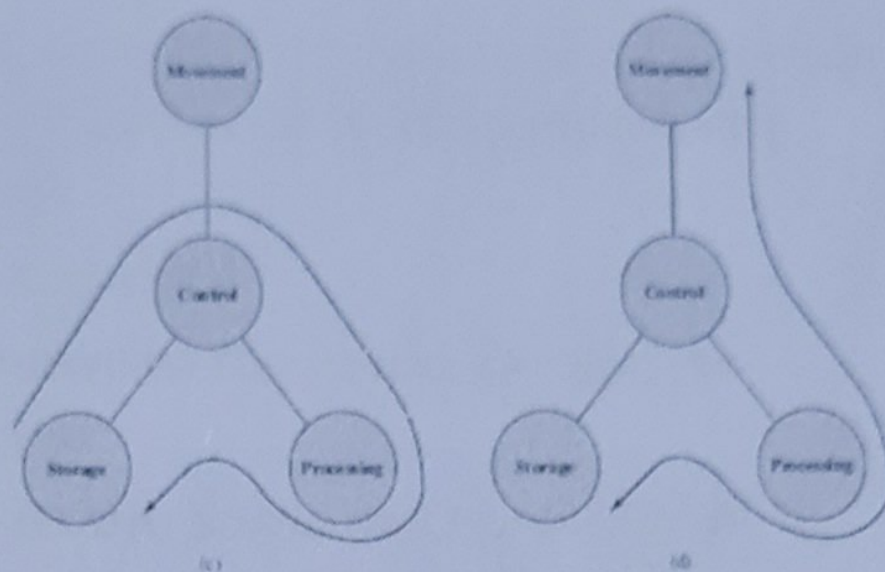


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Possible Computer Operation

The final two diagrams show operations involving **data processing**, on data either in storage (Figure c) or en route between storage and the external environment (Figure d)



البيانات
معالجة
داخل



Structure of the Computer



- The most prominent elements on the motherboard are the **chips**.
- A **chip** is a single piece of semiconducting material, typically **silicon**, upon which electronic circuits and logic gates are fabricated.
- The resulting product is referred to as a **Integrated Circuit (IC)** .
- The motherboard contains a slot or socket for the processor chip, which typically contains multiple individual cores, in what is known as a **multicore processor**.
- There are also slots for **memory chips, I/O controller chips, and other key computer components**